

Advanced (Habanero)

- Q1.** What is the purpose of a two-way frequency table?
- To display one categorical variable
 - To compare two numerical variables
 - To analyze the relationship between two categorical variables
 - To calculate probabilities
 - To graph data
- Q2.** A survey of 20 students finds that 8 students like pizza and 12 students like sushi. If 4 students like both, what is the number of students who like only pizza?
- 4
 - 6
 - 8
 - 12
 - 16
- Q3.** Given a two-way table with rows representing gender and columns representing favorite sport, which of the following statements is supported by the data?
- More males like basketball than females
 - More females like soccer than males
 - The majority of students like basketball
 - The data does not support any conclusions about gender and favorite sport
 - More students like soccer than basketball
- Q4.** If $P(A)$ is the probability of event A occurring and $P(B)$ is the probability of event B occurring, what is $P(A \text{ or } B)$ if A and B are mutually exclusive?
- $P(A) + P(B)$
 - $P(A) - P(B)$
 - $P(A) \cdot P(B)$
 - $P(A)/P(B)$
 - $1 - P(A)$
- Q5.** A two-way table shows the relationship between the number of hours studied and the grade received on a test. Which of the following can be concluded from the data?
- Studying more hours results in a higher grade
 - There is no relationship between hours studied and grade received
 - The data is insufficient to make a conclusion
 - The number of hours studied has no effect on the grade
 - More students received a higher grade than a lower grade
- Q6.** What is the relative frequency of students who like only pizza?
- | | |
|--------|-----|
| rac15 | (A) |
| rac14 | (A) |
| rac310 | (A) |
| rac25 | (A) |
| rac12 | (A) |
- Q7.** If a two-way table has 3 rows and 4 columns, how many cells are in the table?
- 6
 - 8
 - 10
 - 12
 - 16
- Q8.** What is the purpose of calculating relative frequencies in a two-way table?
- To compare the number of students in each category
 - To analyze the relationship between two categorical variables
 - To determine the probability of an event occurring
 - To identify patterns and trends in the data
 - To create a graph or chart

Open Ended Questions (FRQ)

$hline_{extMale}$	ext_{Pizza}	8
ext_{Female}		6
$hline_{extTotal}$		14
$endarray$		

- Q9.** Using the two-way table below, calculate the relative frequency of students who like both pizza and sushi.

- Q9.** Explain how to analyze the ass

Chef's Tips

- Tip 1: Ensure students understand the concept of relative frequency and how to calculate it
- Tip 2: Provide opportunities for students to practice analyzing two-way tables and identifying patterns and trends
- Tip 3: Encourage students to use real-world examples to illustrate their understanding of two-way tables and relative frequencies